

C++ Stream Classes

What is Stream?

- A stream is an abstraction. It is a sequence of bytes.
- It represents a device on which input and output operations are performed.
- It can be represented as a source or destination of characters of indefinite length.
- It is generally associated to a physical source or destination of characters like a disk file, keyboard or console.
- C++ provides standard **iostream** library to operate with streams.
- The **iostream** is an object-oriented library which provides Input/Output functionality using streams.

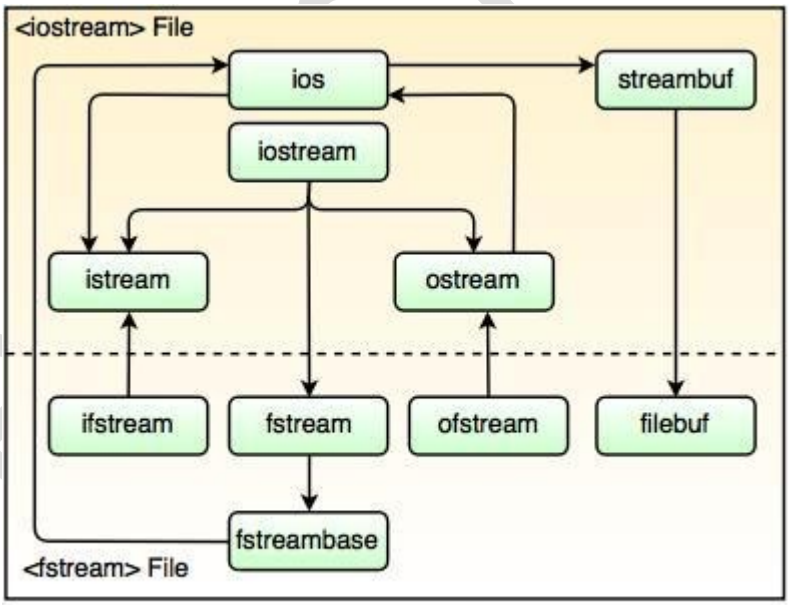
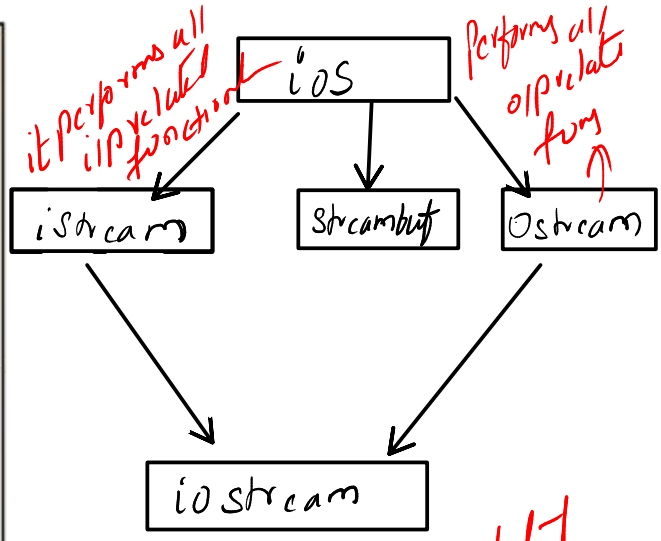


Fig. Stream Class Hierarchy



In the above figure, **ios** is the base class. The **iostream** class is derived from **istream** and **ostream** classes. The **ifstream** and **ofstream** are derived from **istream** and **ostream**, respectively. These classes handles input and

output with the disk files.

The **fstream.h** header file contains a declaration of **ifstream, ofstream** and **fstream** classes. The **iostream.h** file contains **istream, ostream** and **iostream** classes and included in the program while doing disk I/O operations.

There are mainly two types of consol IO operations.

- 1. Unformatted consol IO -
- 2. Formatted consol IO

Unformatted console IO operations

We use the following built-in functions to perform operations of type unformatted consol IO operations.

get( ) and put( )

The get( ) is a method of cin object used to input a single character from the standard input device (keyboard). Its main property is that it allows wide spaces and newline character. The get() function does not have any return value (void get( )).

The put() is a method of cout object and it is used to print the specified character on standard output device (monitor).

Example :

```
#include <iostream>
using namespace std;
int main()
{
      char ch;
      cout<<"press any key:"<<endl;
      cin.get(ch);
      cout<<"You have pressed:";
      cout.put(ch);
      return 0;
}
Output :
      press any key: A
      You have pressed: A
```

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getline(char *buffer,int size) and write(char * buffer, int n)

The getline(char *buffer,int size) is a method of cin object and it is used to input a string with multiple spaces.

The write(char * buffer, int n) is a method of cout object and it is used to read n character from buffer variable.

```
Example :#include<iostream>
#include<cstring>
using namespace std;
int main()
{
     char ch[20];
     cout<<"enter the string:"<<endl;
     cin.getline(ch,20);
     cout.write(ch,strlen(ch));
     return 0;
}
Output :
enter the string:
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```

cin and cout objects

Cin:

The cin is an object of class istream. It is used to accept the input from the standard input device i.e. keyboard. It is associated with the standard C input stream stdin.

cin declaration
extern istream cin;

It is defined in <iostream> header file.

The cin object is used along with the extraction operator (>>) in order to receive a stream of characters.

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`cout`

The cout object in C++ is an object of class ostream. It is used to display the output to the standard output device i.e. monitor. It is associated with the standard C output stream stdout.

cout declaration

`extern ostream cout;`
It is defined in iostream header file.

The `cout` object is used along with the insertion operator (<<) in order to display a stream of characters.

Formatted console IO operations

1. width(): The width method is used to set the required field width. The output will be displayed in the given width

2. precision(): The precision method is used to set the number of the decimal point to a float value
3. fill(): The fill method is used to set a character to fill in the blank space of a field
4. setf(): The setf method is used to set various flags for formatting output
5. unsetf(): The unsetf method is used To remove the flag setting.

Example

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
        int x=20;
        cout.width(10);
        cout<<x<<endl;
        cout.width(10);
        cout.fill('*');
        return 0;
}
```

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Output :

```
        20
********20
```

Formatting IO operations using Manipulators :

The C++ programming language provides the following built-in functions to display the output in formatted form. These built-in functions are available in the header file iomanip.h.

setw(int) and setfill(char)

The setw(int) is a method used to set width of the output.

The setfill(char) is a method used to fill specified character at unused space.

Example

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
        int x=20;
        cout<<setw(10);
        cout<<x<<endl;
        cout<<setw(10)<<setfill('*')<<x<<endl;
        return 0;
}
```

Output :

```
        20
********20
```

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